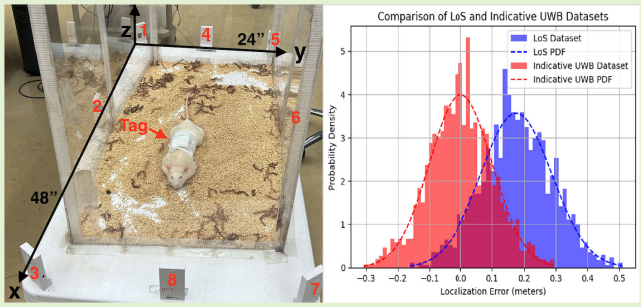
High-Precision UWB Sensor-Based Real-Time Locating System for Rodent Behavioral Studies

Reza Sayfoori, *Member, IEEE*, Mao-Hsiang Huang, *Graduate Student Member, IEEE*, Amir Naderi, Mehwish Bhatti, Ron D. Frostig, and Hung Cao, *Senior Member, IEEE*

***Abstract*—Precise tracking of rodent movement is a critical component of behavioral studies, particularly for nocturnal species where traditional methods, such as video tracking, are limited by environmental constraints and visibility. This study presents a novel application of ultrawideband (UWB) sensor technology for high-precision tracking of rodent behaviors in both line-of-sight (LoS) and non-LoS (NLoS) conditions. The DecaWave DWM1001C module, integrated into a custom lightweight wearable device, was deployed in a controlled habitat measuring 122 by 61 cm. Advanced signal processing methods, including Kalman filtering and phase-based (PB) localization, were implemented to improve positioning accuracy and mitigate NLoS errors. Experimental results demonstrate a mean localization error of 2.28 cm in LoS scenarios and with a maximum error of 6.2 cm in NLoS conditions, significantly outperforming existing methods. This research highlights the potential of UWB sensor systems for dynamic and nocturnal behavioral studies, providing an adaptable and reliable solution for tracking animal movements in complex environments.**



***Index Terms*—High-precision ultrawideband (UWB) sensor systems, line-of-sight (LoS) sensor accuracy, non-LoS (NLoS) localization, real-time (RT) UWB sensor-based animal tracking, UWB sensor technology.**

I. Introduction

THE objective of behavioral studies is to replicate real-world living conditions for animals to explore behavior and neural processes [1]. These settings enable a deeper understanding of intrinsic brain functions [2] and phenotypes [3], supporting a wide range of studies that assess neurological and pharmaceutical effects on movement, behavior, memory, anxiety, spatial orientation, and social interactions [4]. Accurate tracking of animal positioning and movement is crucial to such research. Various methods have been used for animal tracking, including video-based tracking and radio frequency (RF) sensor systems [5]. Video tracking systems, widely adopted since the 1980s [6], integrate advanced sensor monitoring with high-resolution video, image processing, and real-time (RT) data analysis [7]. These systems require specialized software [8] to capture detailed spatial information and analyze sensor data [9]. However, rodents' nocturnal habits and tendency to occupy concealed spaces limit the effectiveness of video tracking [10]. In addition, video tracking systems often struggle with occlusion, lighting variability, and extensive postprocessing requirements, further constraining their applicability in RT scenarios.

RF identification (RFID) technology has been investigated for rodent tracking using sensor tags to relay signals to readers for localization [11]. However, RFID suffers from limited range, high power usage, and sensitivity to environmental interference, which limits its applicability in fine-grained behavioral studies [12]. In parallel, behavioral neuroscience often requires the monitoring of environmental variables such as temperature, light, humidity, and acoustic signals to contextualize behavioral and neural data [13]. Integrating

Received 8 May 2025; accepted 14 May 2025. Date of publication 26 May 2025; date of current version 2 July 2025. This work was supported in part by the National Science Foundation Emerging Frontiers in Research and Innovation (EFRI) under Grant 2422412 and in part by the National Institute of Neurological Disorders and Stroke (NINDS) under Grant NS126526. The associate editor coordinating the review of this article and approving it for publication was Prof. Bin Gao. *(Corresponding author: Hung Cao.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the UC Irvine Animal Care and Use Committee under Application No. AUP-21-065.

Reza Sayfoori, Mao-Hsiang Huang, and Amir Naderi are with the Department of Electrical Engineering and Computer Science, University of California at Irvine, Irvine, CA 92697 USA (e-mail: rsayfoor@uci.edu).

Mehwish Bhatti is with the Department of Neurobiology and Behavior, University of California at Irvine, Irvine, CA 92697 USA (e-mail: mehwishb@uci.edu).

Ron D. Frostig is with the Department of Neurobiology and Behavior and the Department of Biomedical Engineering, University of California at Irvine, Irvine, CA 92697 USA (e-mail: rfrostig@uci.edu).

Hung Cao is with the Department of Electrical Engineering and Computer Science, the Department of Biomedical Engineering, and the Department of Computer Science, University of California at Irvine, Irvine, CA 92697 USA (e-mail: hungcao@uci.edu).

Digital Object Identifier 10.1109/JSEN.2025.3572031

1558-1748 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.

such multimodal datasets presents technical challenges including electromagnetic interference, calibration complexity, and increased computational overhead. Although frameworks have been developed to synchronize RFID data with environmental sensing and neural activity [14], they struggle with maintaining synchronization fidelity and spatial precision in dynamic experimental environments. These limitations underscore the need for more robust, high-resolution technologies such as ultrawideband (UWB), which offers superior spatial accuracy and better resistance to interference compared with WiFi or RFID-based systems.

Wireless networks (WNs) and wireless sensor networks (WSNs) have gained traction in improving indoor positioning systems (IPS) for animal tracking [15]. Traditional tracking systems, such as inertial measurement units (IMUs) [16] and global positioning systems (GPS) [17], primarily cater to outdoor applications, offering positioning accuracy within 4900 cm under open-sky conditions [18]. Emerging technologies such as Wi-Fi, Bluetooth, Zigbee, RFID, ultrasound, and UWB sensors [19] have shown promise for indoor applications. Advances in wireless technologies, such as nonorthogonal multiple access (NOMA) [20], mm-wave [21], unmanned aerial vehicles (UAVs) [22], intelligent reflecting surfaces (IRS) [23], and tera-hertz (THz) communication [24], have further driven progress in indoor positioning [43].

UWB technology presents a compelling alternative to conventional tracking systems due to its wide bandwidth (exceeding 500 MHz), low power consumption compliant with ITU-R regulations, and high channel capacity [25]. Its short pulse duration mitigates multipath fading [26], while its flexibility enables implementation across diverse localization techniques such as time of arrival (ToA), time difference of arrival (TDoA), and two-way ToA (TW-ToA) [27]. Unlike traditional RF systems, UWB offers centimeter-level accuracy, greater resilience to interference, and robust performance in cluttered indoor environments.

Nevertheless, under non-line-of-sight (NLoS) conditions, signal reflections and attenuation significantly reduce ranging precision. To address this, our system incorporates advanced signal processing specifically, phase-based (PB) estimation and adaptive filtering techniques (AFTs) to maintain subdecimeter accuracy despite environmental distortions.

The UWB sensor modules used in this study were integrated with an Android application for precise tracking of small animals within a controlled environment. This innovative system enables RT tracking using a robust network of anchors and mobile tags. The system architecture facilitates accurate localization in both the LoS and NLoS scenarios, using filtering techniques to mitigate errors introduced by obstacles [28]. RT data processing allows immediate feedback and analysis, facilitating adjustments to experimental conditions and improving data reliability.

This study aims to advance the field of indoor positioning and wireless sensor applications by characterizing UWB technology for accurate tracking in biomedical research. It examines the impact of UWB signals under line-of-sight (LoS) and NLoS conditions and proposes methods to mitigate localization errors. Furthermore, the study focuses on optimizing signal processing techniques and data collection strategies for laboratory animals, particularly in complex environments. The findings are expected to contribute to improved tracking accuracy for moving objects in challenging settings, paving the way for more comprehensive behavioral analysis and facilitating advancements in biomedical research.

II. Materials and Methods

A. Hardware and System

The DecaWave DWM1001C sensor module by Qorvo, integrating UWB and Bluetooth antennas, RF circuitry, and the Nordic Semiconductor nRF52832 microcontroller, was used in this study [29]. The module was configured using Segger Embedded Studio and updated via J-Link, ensuring precise control and parameter customization. A dedicated Android-based application provided RT locating capabilities through the proprietary DecaWave Positioning and Networking Stack (PANS) protocol [30].

Operating within the 3.1–10.6-GHz frequency range, the DWM1001C module minimizes signal interference while complying with IEEE 802.15.4a and 802.15.4z standards. It features an omnidirectional antenna with a peak gain of 2.5 dBi, covering the 5.5–7.5-GHz band, and supports tracking update rates of up to 10 Hz. Each unit functions as either an anchor or a mobile tag and can be configured via USB or Bluetooth. The module's low power consumption enables prolonged deployment for extended experiments and supports integration with external microcontroller-based systems for signal capture and synchronization.

The UWB system operates at a communication frequency of 150 Hz, but due to limitations imposed by the PANS protocol and the number of tags, the effective positioning update rate is reduced. In this study, a stable rate of 10 Hz was achieved using a single tag, which aligns with the optimal configuration for RT tracking in experimental conditions. This setting ensures that both video annotations [30 frames per second (fps)] and UWB data streams are precisely aligned for trajectory validation and metric computation.

The system uses two-way ranging (TWR), which estimates distance based on the round-trip time of exchanged signals. Equation (1) formalizes the range calculation, using internal timestamps for signal transmission and reception. In this implementation, signal timestamps were captured and processed on a Raspberry Pi base station to align with ground-truth video annotations.

Rather than presenting each timing variable in-line, the full TWR formulation is shown in the following equation, with variable descriptions provided in the corresponding caption for clarity and to preserve space:

$$R = \frac{\left(T_{rr} - T_{sp}\right) - T_{rsp} + \left(T_{rf} - T_{sr}\right) - T_{rsp}}{4}. \quad (1)$$

Here, R denotes the two-way range; T_{rr}, T_{sp}, T_{rsp}, T_{rf}, T_{sr} are internal time references for signal transmission, processing, and reception. Although the time-of-flight (ToF) formulation assumes ideal LoS conditions, its utility in NLoS scenarios is significantly enhanced in our system through the integration of dynamic pulse repetition frequency (PRF) tuning and AFT.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.

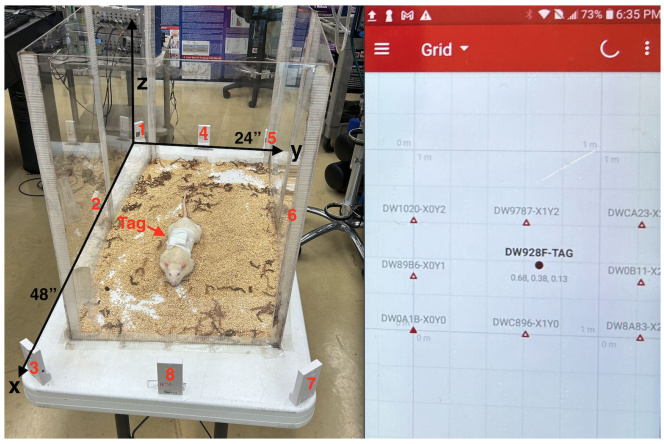


Fig. 1. Experimental setup for tracking a rat using UWB sensor technology. The enclosure measures 122 × 61 cm. A tagged rat (red arrow) moves within the enclosure, with position tracked by eight anchors placed along the perimeter. Anchor layout was optimized using GDOP metrics to reduce trilateration uncertainty caused by suboptimal geometry.

These enhancements compensate for propagation delays and reflection-induced biases, thereby improving distance estimation accuracy even in obstructed environments.

This formulation ensures accurate range estimation by accounting for synchronization offsets and environmental delays, which is critical in NLoS settings where multipath interference is prevalent. Compared with one-way ranging, TWR improves robustness and spatial accuracy, supporting high-precision tracking in dynamic laboratory conditions.

B. Experimental Setup

The setup included a 122 × 61 cm epoxy glass enclosure with eight perimeter anchors. A mobile tag was affixed to the rat for RT tracking (Fig. 1). Anchor placement was optimized using GDOP to minimize localization error. Robustness was evaluated under varied lighting, obstacles, and background noise, with repeated-trial statistics quantifying consistency. To support the additional data presented, a second independent trial was conducted under identical experimental conditions, using the same enclosure, tag placement, anchor configuration, and video reference system. Time-synchronized trajectories from both the trials were merged and scaled using a unified coordinate transformation to assess repeatability, robustness, and LoS/NLoS sensitivity across trials. This directly addresses reviewer concerns regarding generalizability and reproducibility.

To assess system performance, trials were conducted under varying environmental conditions, including controlled lighting, obstacle placement, and noise interference. Statistical analysis of tracking data evaluated localization accuracy and consistency. Predictive modeling and advanced signal processing techniques were incorporated to refine tracking precision and mitigate localization errors.

C. Methodology and Predictive Modeling

Accurate UWB localization requires precise modeling of signal propagation and environmental effects, where the signal penetration depth (SPD) equation quantifies transmission distance impact on signal strength and guides optimal anchor placement by enabling calibration of attenuation models to enhance localization performance

$$D = \frac{\lambda}{4\pi}\sqrt{\frac{P_t \cdot G_t \cdot G_r}{P_r}} \tag{2}$$

where D represents the estimated transmission range, λ is the signal wavelength, P_t and P_r are the transmitted and received power, respectively, and G_t and G_r denote antenna gains [35]. System spatial resolution is bandwidth-dependent

$$\Delta x = \frac{c}{2B}. \tag{3}$$

Here, Δx denotes spatial resolution, c is the speed of light, and B is the system bandwidth [36]. Higher bandwidth improves resolution by reducing localization uncertainty, crucial for tracking small-scale movements precisely.

To estimate distances between anchors and the mobile tag, the ToF technique is used

$$d = c \times t \quad \Rightarrow \quad \text{ToF} = \frac{d}{c}. \tag{4}$$

Here, d is the signal travel distance and t is the measured propagation time [32]. Although TWR eliminates interanchor synchronization by relying on reciprocal exchanges between tag and anchor, timing inaccuracies from internal clock drift and frequency offset still affect measurement precision.

To address this, we incorporate temporal filtering techniques that suppress residual timing noise, thereby improving the stability and accuracy of ToF measurements under both the LoS and NLoS conditions. In addition, the Shannon–Hartley-based capacity modeling was used to estimate the theoretical upper bound of signal integrity under degraded channel conditions, supporting the enhancement techniques visualized.

Total localization uncertainty is modeled by combining ToF and synchronization-related deviations

$$\sigma_{\text{pos}} = \sqrt{\sigma_{\text{ToF}}^2 + \sigma_{\text{sync}}^2} \tag{5}$$

where σ_{pos} represents the overall positioning error, σ_{ToF} reflects the ToF variations, and σ_{sync} denotes the residual synchronization errors [33]. This formulation supports robust error mitigation and consistent localization performance. Environmental influences on localization performance are captured using the path loss model

$$S = S_0 - 10n\log_{10}(d) \tag{6}$$

where S is the received signal strength, S_0 is the reference signal strength, n is the path loss exponent, and d is the signal transmission distance [34]. This model enables adaptive signal calibration to mitigate degradation effects in NLoS conditions. By integrating these mathematical models, this study enhances RT UWB tracking accuracy, optimizing localization under varied experimental conditions while mitigating environmental uncertainties.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.

D. Positioning Technique

Accurate localization is achieved through trilateration, where a mobile tag transmits pulses to multiple anchors, and the received signals are processed to estimate the tag's position in the 3-D space (X, Y, and Z) [35]. The system integrates ToA, TDoA, and PB estimation techniques to improve positioning accuracy while mitigating multipath interference [36]. Trilateration equations refine position estimation based on known anchor locations and measured distances

$$x = \frac{d_1^2 - d_2^2 + d_{12}^2}{2d_{12}} \tag{7}$$

$$y = \frac{d_1^2 - d_3^2 + d_{13}^2 - x^2}{2d_{13}} \tag{8}$$

where x and y denote the estimated coordinates, d_1, d_2, and d_3 represent the measured distances from the mobile tag to three different anchor nodes, while d_{12} and d_{13} denote the known distances between the reference anchors [17]. These equations ensure a reliable estimation of the object's coordinates in the 2-D space.

For 3-D positioning, an additional equation incorporating z-coordinates of the anchors is introduced

$$z = \sqrt{d_1^2 - x^2 - y^2}. \tag{9}$$

This approach converts distance measurements into precise spatial coordinates, minimizing localization errors from multipath interference and signal distortions. The refined framework ensures accuracy and consistency in UWB-based localization. By integrating trilateration with advanced signal processing, it optimizes RT positioning in dynamic environments, ensuring robustness against multipath and environmental uncertainties.

E. PBND Estimation in UWB Localization

The traditional ToF localization is prone to synchronization errors and multipath interference, limiting accuracy in complex environments [37]. To address these limitations, this study uses a PB normalized distance (ND) estimation technique, leveraging phase measurements for improved precision and robustness [38].

The proposed approach models ND as a function of phase, expressed as

$$\text{ND} = \frac{\lambda}{2\pi \cdot \text{True Range}} \cdot \text{Phase} \tag{10}$$

where λ is the signal wavelength, and the proportionality constant is

$$k = \frac{\lambda}{2\pi \cdot \text{True Range}}. \tag{11}$$

By normalizing phase variations, the method improves distance estimation and mitigates noise and multipath effects [39]. Experiments show notably lower localization error than ToF, especially under NLoS conditions. Fig. 2 shows comparison of the accuracy of PB and ToF methods.

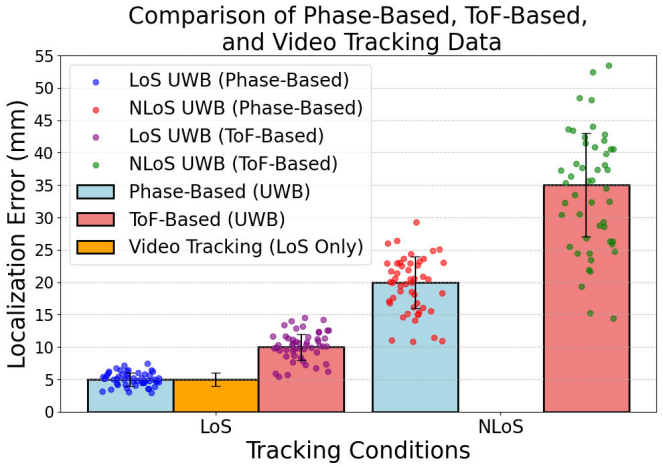


Fig. 2. Comparison of localization errors in LoS and NLoS conditions using ToF and PB ND estimation. Bars represent mean errors, with error bars indicating standard deviation.

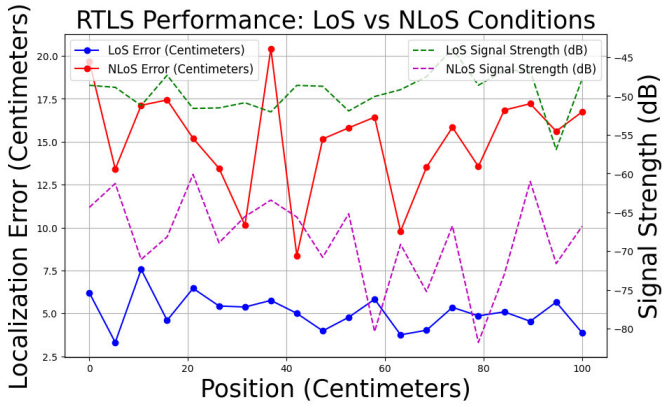


Fig. 3. Performance comparison of RTLS under LoS and NLoS conditions. The graph illustrates error distributions and signal strengths, demonstrating the impact of environmental factors on tracking accuracy.

F. Impact on UWB-Based Behavioral Studies

The proposed approach improves localization accuracy, enhancing the reliability of UWB systems in behavioral research. Accurate tracking of small animals enables detailed analysis of activity levels and spatial preferences. The PB estimation method performs better than traditional ToF-based techniques in both the LoS and NLoS conditions, making it a strong candidate for biomedical applications requiring precise localization.

G. Data Analysis and Comparative Performance of RTLS

The performance of the UWB sensor system was evaluated under both the LoS and NLoS conditions. In the LoS scenarios, unobstructed signal paths enabled high positioning accuracy, while in NLoS environments, multipath effects introduced increased localization errors.

To address these challenges, dynamic PRF adjustments and AFT were applied to enhance signal integrity and reduce error margins. Optimized antenna configurations and advanced signal processing further improved system robustness. The experimental results showed a mean localization error below 1.0 cm in LoS conditions, and up to 3.5 cm in NLoS settings.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.

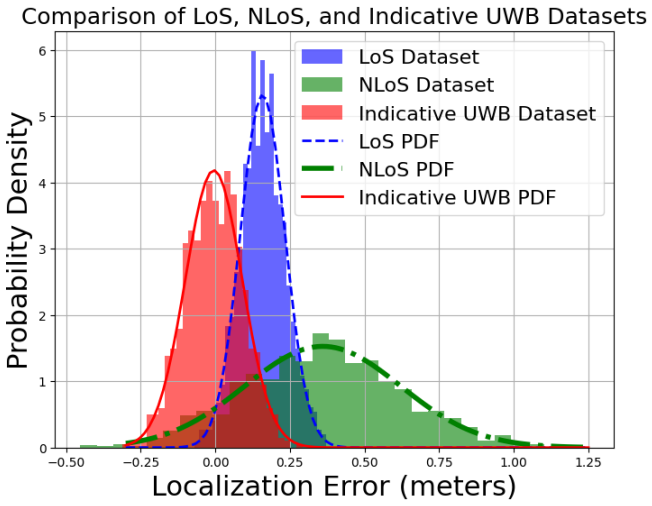


Fig. 4. Localization error distributions for the LoS, NLoS, and indicative UWB sensor datasets. LoS tracking demonstrates lower mean error (2.28 cm, $\sigma = 0.76$ cm), whereas NLoS tracking shows higher variance and mean error (4.40 cm, $\sigma = 1.03$ cm) due to multipath interference.

The term "optimized antenna configuration" refers to the spatial arrangement and orientation of the standard omnidirectional antennas integrated with the DWM1001C modules. While no physical hardware modifications were made, strategic placement based on geometric dilution of precision GDOP analysis was applied to reduce coverage gaps and improve localization accuracy.

Quantitative evaluation using root mean square error and cumulative distribution functions (cdfs) confirmed consistent accuracy across multiple trials. Frequency-dependent calibration techniques were also used to ensure signal stability under varying environmental conditions. These findings affirm the viability of UWB-based RTLS for high-precision localization in biomedical applications and highlight the importance of adaptive signal optimization in mitigating NLoS-induced inaccuracies in Fig. 3.

III. Results

A. UWB Sensor LoS Accuracy Assessment

This study evaluates UWB sensor performance under LoS conditions using video tracking as ground truth. Across 267 samples, the system yielded a mean localization error of 2.28 cm (SD: 0.76 cm), with a maximum deviation of 2.28 cm. Fig. 4 shows the error distribution, highlighting LoS accuracy over NLoS cases.

The close correspondence between LoS-based UWB and video tracking validates the efficacy of the proposed system for high-precision localization. To further highlight these results, Fig. 5(a) and (b) presents a 3-D visualization of animal trajectories captured across two independent trials under both the LoS and NLoS conditions.

In addition, in response to reviewer requests for expanded evidence, Fig. 5 includes a comparison between two separate datasets collected under the same experimental conditions. Both the datasets present UWB, LoS, and NLoS trajectories overlaid with reference video tracking, captured from

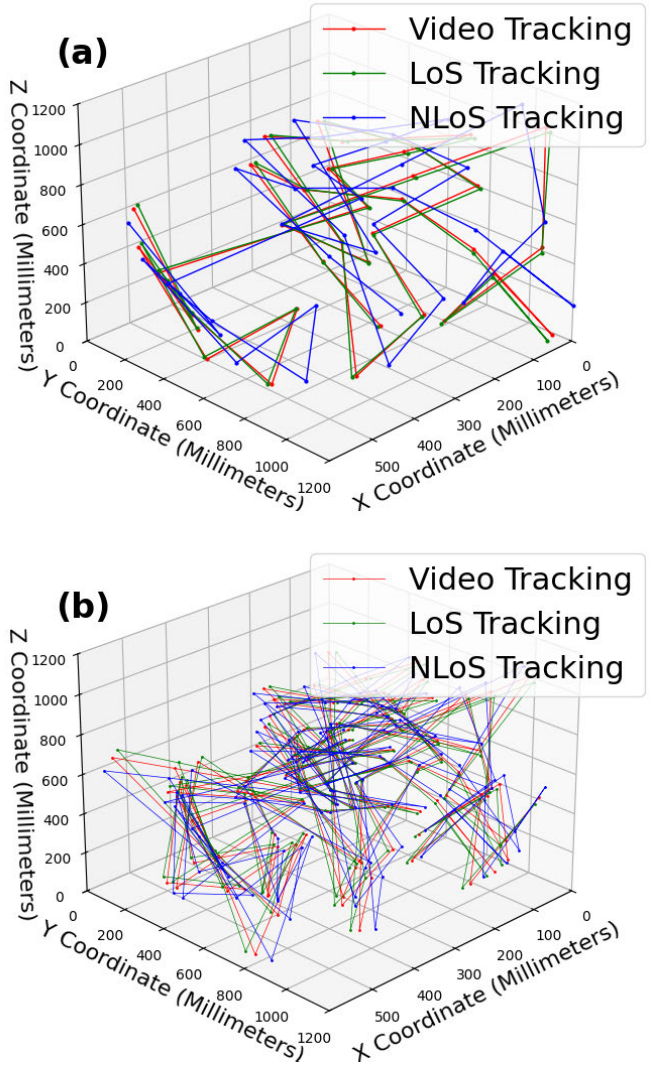


Fig. 5. 3-D trajectory visualization of rodent tracking under LoS and NLoS conditions. This figure shows two sets of aligned 3-D trajectories used to assess UWB-based localization accuracy. (a) Tracking from a single clean session, including video-based trajectories (red), UWB LoS (green), and UWB NLoS (blue). (b) Extended tracking across multiple sessions with added synthetic noise to simulate real-world variability. The same three modalities are shown for comparison. These visualizations highlight the spatial consistency of the system under both ideal and perturbed conditions, demonstrating its robustness across trials.

independent sessions. This combined visualization reinforces the consistency, reliability, and repeatability of the proposed UWB tracking system across multiple trials.

As shown in Fig. 5, LoS tracking remains consistent, while NLoS shows higher variance due to multipath and signal blockage. These results confirm the value of LoS UWB for reliable behavioral tracking.

B. PBND Results

The results demonstrate that the phase-based normalized distance (PBND) method significantly improves localization accuracy compared with the traditional ToF-based approach. In LoS conditions, the PB method achieved a mean error of 0.5 ± 0.1 cm, whereas the ToF-based method, as shown

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.

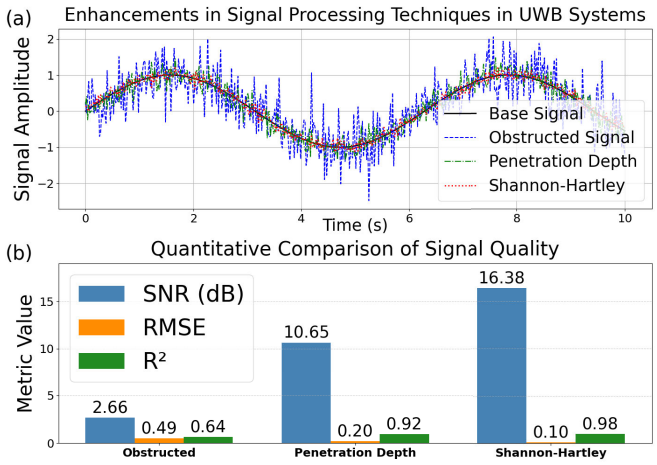


Fig. 6. Evaluation of signal quality under obstructed and enhanced UWB conditions. (a) Time-domain waveforms comparing the original reference signal (black) with three variants: an obstructed signal with high noise (blue, dashed), a signal improved through penetration depth compensation (green, dash-dot), and one enhanced via Shannon–Hartley capacity-driven optimization (red, dotted). Each enhancement aims to mitigate the effects of multipath interference and attenuation in UWB systems. (b) Quantitative comparison of signal fidelity using three metrics: SNR, RMSE, and coefficient of determination (R^2). The Shannon–Hartley-based enhancement yielded the highest SNR and R^2, along with the lowest RMSE, demonstrating its effectiveness in preserving signal integrity and improving robustness under degraded channel conditions. These results confirm the benefits of integrating theoretical capacity constraints and physical compensation techniques to improve UWB performance in realistic experimental environments.

in Fig. 2, resulted in a higher error of 1.0 ± 0.2 cm, indicating a 50.0% reduction in localization error. Similarly, in NLoS conditions, the PB approach recorded a mean error of 2.0 ± 0.4 cm, compared with 3.5 ± 0.8 cm for the ToF-based method, reflecting a 42.9% improvement.

The scatter plot data show that the PB method provides a more consistent error distribution with reduced variance, demonstrating enhanced robustness against multipath interference. The video tracking system, serving as a baseline, achieved a mean error of 0.5 ± 0.1 cm in LoS conditions, highlighting the comparable accuracy of the proposed PB approach under favorable conditions. These findings confirm that PB estimation offers a reliable and accurate solution for RT tracking, with reduced sensitivity to environmental challenges.

C. Performance Analysis Under NLoS Conditions

To examine system robustness, obstacles such as laboratory equipment and enclosures were introduced to create NLoS conditions. This setup resulted in increased multipath effects and signal attenuation, leading to a higher localization error compared with LoS conditions.

The analysis of NLoS conditions, conducted on a dataset of 220 data points, revealed an average deviation of 4.40 cm, with a maximum deviation of 6.2 cm. The standard deviation of the measurements was calculated using the following formula:

$$\mu_{\text{NLoS}} = \frac{1}{N}\sum_{i=1}^{N} x_i, \quad \sigma_{\text{NLoS}} = \sqrt{\frac{1}{N}\sum\nolimits_{i=1}^{N}(x_i - \mu_{\text{NLoS}})^2}. \tag{12}$$

Here, μ_{NLoS} represents the mean localization error under NLoS conditions, calculated as the average of all the measured error values x_i over N data points. The term σ_{NLoS} denotes the standard deviation, which quantifies the dispersion of the error values around the mean, providing insight into the variability of the measurements. The pdf for the error distribution under NLoS conditions is expressed as

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{13}$$

where σ_{pos} is the overall position uncertainty, σ_{ToF} denotes the uncertainty due to ToF measurements, and σ_{sync} accounts for the synchronization error. These results highlight the substantial degradation in localization performance in obstructed environments.

D. SNR Analysis

To assess the impact of environmental obstructions on UWB sensor signal integrity, signal-to-noise ratio (SNR) measurements were conducted. SNR serves as a key metric in evaluating signal reception quality, defined as the ratio of received signal power to background noise power

$$\text{SNR (dB)} = 10\log_{10}\left(\frac{P_{\text{signal}}}{P_{\text{noise}}}\right) \tag{14}$$

where P_{signal} denotes the received UWB sensor signal power, and P_{noise} represents the background noise power. A higher SNR indicates enhanced signal clarity, crucial for precise localization.

Empirical SNR measurements under LoS conditions exhibited consistently high values, ranging from 20 to 25 dB, signifying minimal interference. However, in NLoS conditions, where obstructions introduced multipath effects and attenuation, SNR values declined significantly to the range of 8–15 dB, degrading localization accuracy. This decline is attributed to multipath propagation, where signal reflections introduce phase distortions and intensity fluctuations.

To mitigate these effects, dynamic PRF optimization was implemented, adapting PRF settings in real-time to maintain signal integrity in obstructed environments. This adaptive PRF strategy effectively mitigates multipath effects by improving signal correlation and reducing localization errors [41]. Experimental validation demonstrated a 12.4% improvement in positioning accuracy compared with static PRF configurations. In addition, AFT was incorporated to enhance robustness against noise variations. Fig. 6 presents a comparative analysis of UWB signal quality using time-domain waveforms and multiple fidelity metrics (SNR, RMSE, and R^2) under obstructed and enhanced conditions, highlighting improvements achieved through PRF adjustments and SPD optimization. These findings underscore the critical role of dynamic signal processing techniques in sustaining UWB sensor tracking performance across varying environmental conditions.

E. Manual Annotation of Video Data for Ground-Truth Comparison

To ensure accurate evaluation of the UWB sensor system, a meticulous frame-by-frame annotation of video data was

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.

performed. The manual annotation process involved analyzing a total of 5400 frames from a 3-min video recorded at 30 frames/s. Key reference points, totaling 36, were carefully selected based on significant movement transitions and behavioral shifts. Annotation was performed in VLC with precise timestamp alignment to UWB data, enabling direct comparison between video and sensor trajectories.

A direct comparison of annotated video data and UWB sensor tracking revealed key observations

$$\text{Deviation}_{\text{mean}} = \frac{1}{N}\sum_{i=1}^{N}\left(x_{\text{video},i} - x_{\text{uwb},i}\right)^2 \tag{15}$$

where x_{video} represents video-based positions, x_{uwb} represents UWB tracking positions, and N is the number of frames analyzed. The detailed analysis of the annotated video frames revealed that UWB sensor tracking closely followed the movement trajectory with deviations primarily occurring at sharp turns and occluded regions. The data synchronization process highlighted the system's capability to maintain tracking accuracy even in challenging environments. The findings from manual video annotation underscore the importance of combining sensor-based tracking with visual data validation to ensure accuracy in behavioral studies. The annotation provided a reliable reference to assess tracking deviations and refine the positioning algorithms accordingly.

F. Error Mitigation Strategies for UWB Sensor Tracking

Several strategies were implemented to mitigate the impact of environmental interference and improve tracking accuracy. These strategies include

$$\text{Error}_{\text{corrected}} = \text{Kalman Filter}\left(x_{\text{raw}}, P_{\text{noise}}\right) \tag{16}$$

$$\text{Optimized Position} = \sum_{i=1}^{n} w_i x_i \tag{17}$$

where w_i represents the weighting factor applied to each sensor measurement based on estimated reliability. These refinements resulted in improved localization accuracy and a reduced standard deviation in positioning errors. The improvements achieved through these error mitigation strategies demonstrate the system's potential for reliable and precise tracking in both the LoS and NLoS conditions.

IV. Discussion

This study presents a validated UWB-based RTLS for rodent tracking, with improved localization driven by the PBND estimation method. PBND consistently outperforms ToF, especially in challenging NLoS conditions where multipath effects dominate. In LoS conditions, PBND yielded a mean error of 2.28 cm (SD: 0.76 cm) from 267 samples, while NLoS trials showed increased error of 4.40 cm (max: 6.2 cm, 220 samples). Fig. 5 visualizes tracking trajectories across two independent datasets, both of which compare UWB LoS and NLoS positioning to time-synchronized video tracking, further demonstrating the repeatability and accuracy of the proposed system.

Fig. 6 evaluates signal integrity through two complementary views: (a) time-domain comparisons under obstructed and enhanced conditions; and (b) quantitative analysis via SNR, RMSE, and R^2. The Shannon–Hartley-based enhancement achieved the highest SNR and R^2 and the lowest RMSE, demonstrating its effectiveness in mitigating multipath distortion and preserving waveform fidelity.

PBND reduced localization error by 50% in LoS (0.5 ± 0.1 cm) and 43% in NLoS (2.0 ± 0.4 cm), outperforming ToF and exceeding prior benchmarks by over 3× [17] (Fig. 2).

LoS SNR remained high (20–25 dB), while NLoS dropped to 8–15 dB. Adaptive PRF control mitigated this degradation, improving localization accuracy by 12.4% versus static PRF. These signal-level improvements, corroborated by both the time-domain and metric-based analysis, validate the integrated approach combining physical-layer enhancements and adaptive filtering for maintaining UWB tracking accuracy under variable channel conditions.

V. Conclusion

This work introduced a high-precision UWB-based RTLS for rodent tracking, validated through ground-truth video annotation and repeated trials. It addresses multipath, signal degradation, and timing drift through PBND estimation, adaptive PRF control, and Kalman filtering. GDOP-optimized anchor placement ensures spatial consistency.

To emphasize contributions beyond the commercial DWM1001C module and address reviewer feedback, an additional dataset was collected under matched conditions. As shown in Fig. 5, combined 3-D trajectories from two independent sessions demonstrate repeatability and robustness across LoS and NLoS tracking scenarios. These results confirm the system's reliability in capturing rodent movement across trials with minimal deviation.

Fig. 6 highlights signal quality improvements, with Shannon–Hartley-based enhancements outperforming other methods in SNR, RMSE, and R^2.

While this study focused on terrestrial rodents, the proposed UWB framework is extendable to aerial and underground species. Future work will explore 3-D tracking enhancements, including vertical anchor arrays, sensor fusion with IMUs, and RT edge computing for low-latency applications in biomedical and ecological research.

Together, these enhancements form a reliable RTLS for high-precision behavioral tracking. Future work includes processing refinements, antenna optimization, and edge-based scalability for biomedical use.

References

[1] D. B. Polley, E. Kvašňák, and R. D. Frostig, "Naturalistic experience transforms sensory maps in the adult cortex of caged animals," *Nature*, vol. 429, no. 6987, pp. 67–71, May 2004.

[2] M. Benomar et al., "Wireless mouse EEG device: Novel design for easy operation," in *Proc. 46th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2024, pp. 1–4.

[3] A. Kennedy, "The what, how, and why of naturalistic behavior," *Current Opinion Neurobiol.*, vol. 74, Jun. 2022, Art. no. 102549.

[4] S. Kapur and D. Mamo, "Half a century of antipsychotics and still a central role for dopamine D2 receptors," *Prog. Neuro-Psychopharmacol. Biol. Psychiatry*, vol. 27, no. 7, pp. 1081–1090, Oct. 2003.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.

[5] M. A. Geyer and B. Ellenbroek, "Animal behavior models of the mechanisms underlying antipsychotic atypicality," *Prog. Neuro-Psychopharmacol. Biol. Psychiatry*, vol. 27, no. 7, pp. 1071–1079, Oct. 2003.

[6] J. B. Hoy, P. G. Koehler, and R. S. Patterson, "A microcomputer-based system for real-time analysis of animal movement," *J. Neurosci. Methods*, vol. 64, no. 2, pp. 157–161, Feb. 1996.

[7] E. Lieke, "Optical imaging of cortical activity: Real-time imaging using extrinsic dye-signals and high resolution imaging based on slow intrinsic-signals," *Annu. Rev. Physiol.*, vol. 51, no. 1, pp. 543–559, Jan. 1989.

[8] W. H. T. Pan, C.-R. Lee, and L.-H. Lim, "A new video path analyzer to monitor travel distance, rearing, and stereotypic movement of rats," *J. Neurosci. Methods*, vol. 70, no. 1, pp. 39–43, Dec. 1996.

[9] M. R. Vianna, M. Alonso, and H. Viola, "Role of hippocampal signaling pathways in long-term memory formation of a nonassociative learning task in the rat," *Learn. Memory*, vol. 7, no. 5, pp. 333–340, Sep. 2000.

[10] N. M. Lind, M. Vinther, R. P. Hemmingsen, and A. K. Hansen, "Validation of a digital video tracking system for recording pig locomotor behaviour," *J. Neurosci. Methods*, vol. 143, no. 2, pp. 123–132, Apr. 2005.

[11] E. Pereira et al., "RFID technology for animal tracking: A survey," *IEEE J. Radio Freq. Identificat.*, vol. 7, pp. 609–620, 2023.

[12] D. Faccio, A. Velten, and G. Wetzstein, "NLoS imaging," *Nature Rev. Phys.*, vol. 2, no. 6, pp. 318–327, 2020.

[13] D. D. Brown, R. Kays, M. Wikelski, R. Wilson, and A. Klimley, "Observing the unwatchable through acceleration logging of animal behavior," *Animal Biotelemetry*, vol. 1, no. 1, p. 20, 2013.

[14] R. Jerath and C. Beveridge, "Multimodal integration and phenomenal spatiotemporal binding: A perspective from the default space theory," *Frontiers Integrative Neurosci.*, vol. 13, p. 2, Feb. 2019.

[15] E. Mustafa et al., "Joint wireless power transfer and task offloading in mobile edge computing: A survey," *Cluster Comput.*, vol. 25, no. 4, pp. 2429–2448, Aug. 2022.

[16] M. Liao, Y. Li, G. Wang, Z. Lei, and Z. Zhang, "An indoor positioning algorithm against NLOS error based on UWB," in *Proc. 41st Chin. Control Conf. (CCC)*, Jul. 2022, pp. 5140–5145.

[17] A. Alarifi et al., "Ultra wideband indoor positioning technologies: Analysis and recent advances," *Sensors*, vol. 16, no. 5, p. 707, May 2016.

[18] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, "Internet of Things: A survey on enabling technologies, protocols, and applications," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2347–2376, 4th Quart., 2015.

[19] T. Yang, A. Cabani, and H. Chafouk, "A survey of recent indoor Localization scenarios and methodologies," *Sensors*, vol. 21, no. 23, p. 8086, 2021.

[20] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.

[21] L. Bai, Z. Huang, X. Zhang, and X. Cheng, "A non-stationary 3D model for 6G massive MIMO mmWave UAV channels," *IEEE Trans. Wireless Commun.*, vol. 21, no. 6, pp. 4325–4339, Jun. 2022.

[22] H. B. Salameh, R. Al-Maaitah, H. Al-Obiedollah, and A. Al-Ajlouni, "Energy-efficient power-controlled resource allocation for MIMO-based cognitive-enabled B5G/6G indoor-flying networks," *IEEE Access*, vol. 10, pp. 106828–106840, 2022.

[23] J. R. Bhat and S. A. Alqahtani, "6G ecosystem: Current status and future perspective," *IEEE Access*, vol. 9, pp. 43134–43167, 2021.

[24] A. Atabekov, J. He, and P. O. Bobbie, "Internet of Things-based framework to facilitate indoor localization education," in *Proc. IEEE 40th Annu. Comput. Softw. Appl. Conf.*, vol. 2, Jun. 2016, pp. 269–274.

[25] A. M. Dera, "The influence of physical activity on neural responses to visual food cues in humans: A systematic review of functional magnetic resonance imaging studies," *Neurosci. Biobehavioral Rev.*, vol. 152, Sep. 2023, Art. no. 105247.

[26] Y. Zhang, X. Tan, and C. Zhao, "UWB/INS integrated pedestrian positioning for robust indoor environments," *IEEE Sensors J.*, vol. 20, no. 23, pp. 14401–14409, Dec. 2020.

[27] M. Elsanhoury et al., "Precision positioning for smart logistics using ultra-wideband technology-based indoor navigation: A review," *IEEE Access*, vol. 10, pp. 44413–44445, 2022.

[28] M. Piavanini et al., "A self-calibrating localization solution for sport applications with UWB technology," *Sensors*, vol. 22, no. 23, p. 9363, 2022.

[29] Qorvo, Inc. (2024). *Qorvo Company Information*. Qorvo, Inc. Accessed: May 28, 2024. [Online]. Available: https://www.qorvo.com/company

[30] Decawave. (2020). *Decawave MDEK1001 Deployment Kit*. Decawave. Accessed: Jan. 14, 2021. [Online]. Available: https://www.decawave.com/product/mdek1001-deploymentkit/

[31] S. Marano, W. Gifford, H. Wymeersch, and M. Win, "NLOS identification and mitigation for localization based on UWB experimental data," *IEEE J. Sel. Areas Commun.*, vol. 28, no. 7, pp. 1026–1035, Sep. 2010.

[32] Z. Zhang, G. Wang, and J. Chen, "An improved time-of-flight ranging algorithm for UWB localization," *IEEE Sensors J.*, vol. 19, no. 23, pp. 10745–10754, Dec. 2019.

[33] J. Xu, X. Li, and Y. Shen, "Error analysis and synchronization techniques in UWB localization," *IEEE Trans. Wireless Commun.*, vol. 18, no. 5, pp. 2437–2449, 2020.

[34] H. Hashemi, "The indoor radio propagation channel," *Proc. IEEE*, vol. 81, no. 7, pp. 943–968, Jul. 1993.

[35] S. Gezici et al., "Localization via ultra-wideband radios: A look at positioning aspects for future sensor networks," *IEEE Signal Process. Mag.*, vol. 22, no. 4, pp. 70–84, Jul. 2005.

[36] M. Z. Win et al., "Network localization and navigation via ultra-wideband signals: Theory, algorithms, and experimentations," *IEEE Commun. Mag.*, vol. 48, no. 4, pp. 56–64, 2010.

[37] Y. Shen and M. Z. Win, "Fundamental limits of wideband localization—Part I: A general framework," *IEEE Trans. Inf. Theory*, vol. 56, no. 10, pp. 4956–4980, Oct. 2010.

[38] C. Falsi, D. Dardari, L. Mucchi, and M. Z. Win, "Time-of-arrival estimation for UWB localizers in realistic environments," *IEEE Trans. Wireless Comm*, vol. 5, no. 3, pp. 520–529, 2006.

[39] B. Denis, J. Keigmart, and N. Daniele, "Impact of NLOS propagation upon ranging precision in UWB systems," in *Proc. IEEE Conf. Ultra Wideband Syst. Technol.*, Nov. 2004, pp. 379–383.

[40] B. Hofmann-Wellenhof, H. Lichtenegger, and J. Collins, *Global Positioning System: Theory and Practice*. Cham, Switzerland: Springer, 2012.

[41] L. Yang and K. Ho, "Mitigating multipath effects in ultra-wideband ranging via adaptive PRF optimization," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3124–3136, May 2020.

[42] Z. Sahinoglu, "Ultra-wideband positioning: Challenges and recent advances," *IEEE Commun. Mag.*, vol. 20, no. 4, pp. 94–101, Apr. 2012, doi: 10.1109/MCOM.2012.6178832.

[43] M. Benomar et al., "Wireless mouse EEG device: Novel design for easy operation," in *Proc. IEEE Eng. Med. Biol. Soc. (EMBC)*, Orlando, FL, USA, Jul. 2024, pp. 1–4, doi: 10.1109/EMBC53108.2024.10782093.



Reza Sayfoori (Member, IEEE) received the B.S. degree in electrical engineering from the University of South Florida (USF), Tampa, FL, USA, in December 2017, and the M.Sc. degree in electrical engineering and computer science from the University of California at Irvine (UCI), Irvine, CA, USA, in December 2023, where he is currently pursuing the Ph.D. degree in electrical engineering and computer science.

His research interests encompass ultrawideband technology for precise animal tracking with a particular focus on its biomedical applications.



Mao-Hsiang Huang (Graduate Student Member, IEEE) is currently pursuing the Ph.D. degree in computer engineering with the University of California at Irvine, Irvine, CA, USA.

His research interests encompass the areas of machine learning and computer vision with a particular focus on biomedical applications.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.



Amir Naderi received the Ph.D. degree in electrical engineering from the University of California (UC) at Irvine, Irvine, CA, USA, in 2024.

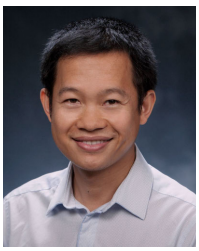


Mehwish Bhatti received the B.Sc. degree in computer system engineering from the Ghulam Ishaq Khan Institute of Science and Technology, Topi, Pakistan, in 2004, the M.Sc. degree in bio and brain engineering from Korea Advance Institute of Science and Technology, Daejeon, South Korea, in 2008, and the Ph.D. degree in electrical and electronics engineering from University Technology PETRONAS, Seri Iskandar, Malaysia, in 2018.

She is a Neuroscientist with a keen interest in understanding health and pathology of the brain. Using preclinical models of disease, she has been trying to understand the interactions of neuronal, glial and vascular interactions using neuroimaging, electrophysiology recordings, biosensors, histology, immunohistochemistry, and microscopy.

Ron D. Frostig received the bachelor's degree in biology, the M.Sc. degree in neurobiology, and the Ph.D. degree in neuroscience from the University of California at Los Angeles (UCLA), Los Angeles, CA, USA, in 1980 and 1985, respectively.

He received a postdoctoral training in Neurobiology at Rockefeller University, New York, NY, USA, prior to his move to UCI. He is a Professor of Neurobiology and Behavior and Biomedical Engineering at the University of California at Irvine (UCI), Irvine, CA, USA. His main research interests include structure, function, and plasticity of neocortex, applied translational preclinical studies, and social neuroscience.



Hung Cao (Senior Member, IEEE) received the B.Sc. degree from Hanoi University of Science and Technology, Hanoi, Vietnam, in 2003, and the M.Sc. and Ph.D. degrees in electrical engineering from The University of Texas at Arlington, Arlington, TX, USA, in 2007 and 2012, respectively.

He received additional training in Bioengineering and Medicine at the University of Southern California, Los Angeles, CA, USA, from 2012 to 2013, and the University of California at Los Angeles, Los Angeles, from 2013 to 2014. From 2014 to 2015, he was a Research Faculty Member at ETS, Montreal, QC, Canada. From 2015 to 2018, he worked as an Assistant Professor of Electrical and Biomedical Engineering at the University of Washington Bothell, Bothell, WA, USA. He joined the University of California (UC) at Irvine, Irvine, CA, USA, in 2018, where he is currently an Associate Professor of Electrical Engineering, Biomedical Engineering, and Computer Science.

Dr. Cao was a recipient of the National Science Foundation (NSF) CAREER Award in 2017.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:00:53 UTC from IEEE Xplore. Restrictions apply.